

Banking & Loan Analytics

Business Problem Statement

A leading bank wants to better understand its loan application and loan portfolio data in order to improve loan approval decisions, customer targeting, and repayment risk monitoring.

The management team wants to understand loan demand across different loan types and customer segments, identify approval and rejection patterns, and detect areas with higher overdue, default, and payment-delay risk.

The analysis will use customer demographics, income, employment type, credit score, loan type, loan amount, approval status, loan status, and payment-delay information to generate actionable business insights.

Overarching Business Question

“How can the bank leverage loan application and portfolio data to understand loan demand, improve approval decisions, identify customer segments, and strengthen repayment risk monitoring?”

Key Business Questions

1. How much loan demand is coming from different loan types?
2. How many loan applications are approved and rejected?
3. Which loan types have the highest application demand?
4. Which customer segments show better approval performance?
5. How does credit score relate to loan approval performance?
6. Which loan types or customer segments have higher overdue and default risk?
7. Where are payment delays concentrated?
8. How can the bank improve loan approval and risk-monitoring decisions?

Deliverables

1. Data Preparation & Modeling — Python

Clean and validate the **100,000-row banking dataset**, check missing values, duplicates, data types, and inconsistent values, and create the required analytical fields.

2. Data Analysis — SQL

Use SQL to perform **30 business-focused queries** covering loan demand, loan types, approval performance, customer segments, repayment risk, and payment delays.

3. Exploratory Data Analysis — Python

Use Pandas and visualization libraries to create **8+ meaningful charts** and identify important patterns and trends in the loan portfolio.

4. Visualization & Insights — Power BI

Build an interactive **4-page Power BI dashboard with 16 KPIs** covering:

- Executive Overview
- Loan Approval Analysis
- Risk & Repayment Analysis
- Customer & Loan Analysis

5. Business Findings & Recommendations

Identify **5+ actual data-based business findings** and provide **3+ actionable recommendations** related to loan approval, customer targeting, and repayment-risk monitoring.

6. Report & Presentation

Prepare clear project documentation covering the business problem, data preparation, analysis methodology, dashboard, findings, recommendations, limitations, and project workflow.